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THE INDUS–TSANGPO SUTURE ZONE IN LADAKH, NORTHWEST HIMALAYA: FURTHER PALAEOMAGNETIC DATA AND IMPLICATIONS *

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ABSTRACT

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A total of 81 samples (244 specimens) from Upper Cretaceous Indus Molasse and Middle to Upper Cretaceous Dras Flyschoids of the Indus–Tsangpo suture zone in Ladakh (northwest Himalaya) has been studied by thermal demagnetization methods.

Both formations showed a characteristic magnetization component indicative for equatorial to low northern palaeolatitudes of acquisition. Similar palaeolatitudes have been obtained before from secondary magnetization components of Early Tertiary age in the Ladakh Intrusives and in the Tibetan Sedimentary Series of central Nepal. The present characteristic components are interpreted likewise as secondary magnetizations which stabilized between 50 and 60 m.y. ago, during Greater India's collision with Asia's southern margin.

The Dras Flyschoids show another magnetic component which, in case of primary origin, indicates acquisition at a low southern palaeolatitude. If correct, this interpretation supports recent suggestions for Late Cretaceous obduction of an island arc on Greater India's northern margin.

INTRODUCTION

Seafloor spreading data show Early Tertiary reductions in the rate of motion of the Indian plate, northwards and relative to Asia (Powell and Conaghan, 1975; Molnar and Tapponnier, 1975; Patriat et al., 1982). These reductions were interpre-

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ted in terms of continental collision between Greater India and southern Asia, at about 55 m.y. ago (Powell and Conaghan, 1975; Powell, 1979). The Indus–Tsangpo suture zone (ITS), representing the relict of this collision, has undergone a complex tectogenesis, which is as yet little understood because of limited geological knowledge of the suture zone and northwards adjacent regions of south-central Asia. Asia's southern margin is made up of an amalgamation of fragments, which broke off from Gondwana during the Early Mesozoic opening of the Neotethys and which collided subsequently with southern Asia during the closure of the Palaeotethys in Middle Mesozoic (Stöcklin, 1977, 1980; Şengör, 1979; Bassoullet et al., 1980a; Tapponnier et al., 1981a) or possibly Early Tertiary times (Leitch, 1982). The Early Tertiary, possibly latest Mesozoic closure of the Neotethys involved collision and obduction of island arcs onto Greater India at an as yet disputed date. The Kohistan Complex and the Dras Volcanics–Dras Flyschoids in the northwest Himalaya can be identified as such (Frank et al., 1977; Klootwijk et al., 1979; Tahirkheli et al., 1979; Bard et al., 1980; Brookfield and Reynolds, 1981; Sinha Roy, 1981; Mattauer et al., 1981; Coward et al., 1982; Honegger et al., 1982). The former extent of the Dras Volcanics–Dras Flyschoids complex may have exceeded its present distribution considerably (Frank et al., 1977). A comparable volcanic sequence exists in the Spongtang Klippe of the Zanskar Mountains (Fuchs, 1977, 1979; Bassoullet et al., 1982: Honegger et al., 1982), and this klippe is analogous in appearance to the exotic nappes of the Amlang La region (Heim and Gansser, 1939; Gansser, 1977). Island arcs quoted to have existed farther east in the Shigatze–Lhasa region (Wu Hao-ruo, 1981; Cao Rong-long, 1981), form most probably part of the Transhimalaya–Gangdeze batholiths of Andean-type (Honegger et al., 1982; Bassoullet et al., 1982), which developed over more than 2000 km at Asia's former southern margin during Late Cretaceous and Early Tertiary time.

Available palaeomagnetic data, notably Early Tertiary secondary magnetization components observed in the Ladakh Intrusives (Klootwijk et al. 1979) and in the Tibetan Sedimentary Series of central Nepal (Klootwijk and Bingham 1980), have been interpreted in support to the above summarized scenario; i.e. in the sense that collision between Greater India and southern Asia or off-shore island arcs occurred 50–60 m.y. ago at equatorial to low northern palaeolatitudes (Klootwijk, 1979, 1981, 1982; Klootwijk and Radhakrishnamurty, 1981). This interpretation, however, needs revision in the light of several important new observations. Namely, evidence for Late Cretaceous tectonic phases in the Himalayan region (Gansser, 1974; Bassoullet et al., 1978; J. Andrieux, pers. commun., 1981; M. Brunel, pers. commun., 1981) and in southern Tibet (Tapponnier et al., 1981a, b; Brunel et al., 1981), and indications for Late Cretaceous obduction of ophiolites or island arc suites onto Greater India (Stoneley, 1974a, b, Bassoullet et al., 1977, 1978, 1980a, 1982; Bard et al., 1980; Brookfield and Reynolds, 1981; Tapponnier et al., 1981a, b). Brookfield and Reynolds (1981) reported a $^{40}Ar/^{39}Ar$ age of 82 ± 6 m.y. for an undeformed hornblende syenite intruded in the metamorphosed ophiolitic melange and Dras

Volcanics at Kargil. They concluded a Coniacian–Santonian age of emplacement, consistent with their interpretation of pre-Maastrichtian emplacement of the related Spongtang Klippe. The latter interpretation is not necessarily contradicted by presence of Lower Tertiary sediment slices at the base of the Spongtang mass. A more serious problem for their interpretation forms the presence of granodiorites (Somau) intruded in the Dras Volcanics near Kargil, dated by U/Pb methods at about 100 m.y. (Schärer, pers. commun., 1982) and 103 ± 3 m.y. (Honegger et al., 1982), whose structural position and chemistry suggest close affinity with the Ladakh Intrusives. In the Kohistan Complex, which is probably related with the Dras Volcanics–Dras Flyschoids complex, radiometric ages of 67 ± 2 m.y. and 84 ± 1.7 m.y. were reported by Bard et al. (1980), Shams (1980) and Desio and Shams (1980) respectively for glaucophane schist situated in a tectonic slice at the base of the Complex in the Shangla region. If representative for obduction on Greater India indeed, these dates place a latest Cretaceous upper limit to the same. In this scenario, the above secondary magnetic components have stabilized during the subsequent—Early Tertiary—suturing between Greater India and south-central Asia.

Indian palaeomagnetic data (Klootwijk, 1979, 1981, 1982; Klootwijk and Radhakrishnamurty, 1981) indicate a southern hemisphere position of Greater India during the Late Cretaceous, with its northern margin at a palaeolatitude of about 20°S. The shape and extension of Asia's southern margin, notably southern Tibet, is as yet ill defined for that period as is the width of the Neotethys oceanic basin between Tibet and Greater India. The distance between both continental blocks may have been far less (Crawford, 1974, 1979; Gansser, 1977, 1981) than shown on most plate tectonic reconstructions. Recent palaeomagnetic data from southern Tibet (Zhu Zhi-wen et al., 1981; Pozzi et al., 1982; Westphal et al., 1983; Courtillot et al., 1982), whose interpretation still awaits further confirmation (Klootwijk 1984), suggest that southern Tibet occupied low to moderate northern latitudes during the Late Cretaceous with Asia rotating slowly southwards (Irving, 1977). If the interpretation of these palaeomagnetic data and the postulated Late Cretacous age of obduction prove to be correct, the obducted island arc and ophiolite sequences must then have originated close to Greater India's northern margin, probably through intra-oceanic subduction (Bard et al., 1980; Tapponnier et al., 1981a, b), at a southern hemisphere palaeolatitude and far off Asia's southern margin.

Control on palinspastic position of intermediary tectonic units is a prerequisite for substantiation and further elaboration of such hypotheses. This formed the main objective for the present reconnaissance-type palaeomagnetic study of the Ladakh region.

REGIONAL GEOLOGY AND SAMPLING

The geology of the Indus region in central Ladakh is described in some detail in several recent studies (Gupta and Kumar, 1975; Shah et al., 1976; Ravi Shankar et

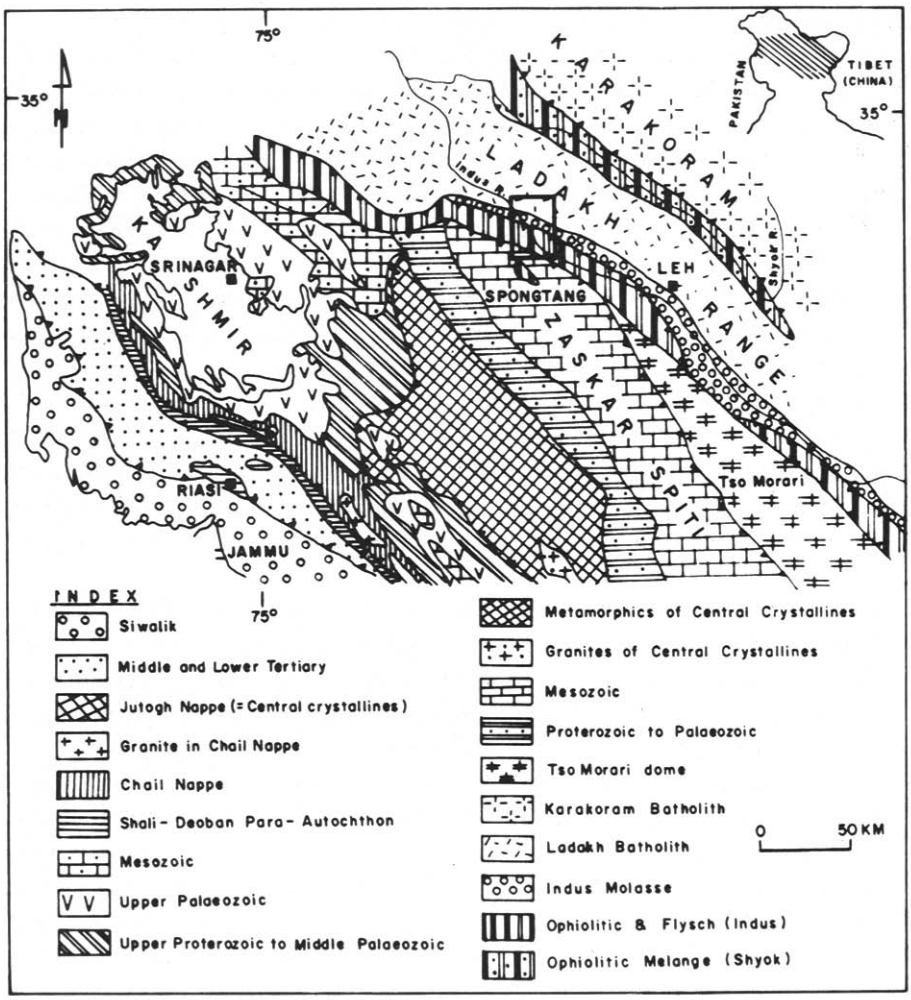


Fig. 1. General geological map of the northwestern Himalaya. Inset corresponds with Fig. 2.

al., 1976; Frank et al., 1977; Andrieux et al., 1977; Pal and Mathur, 1977; Fuchs, 1977, 1979; Bassoullet et al., 1978; 1980a, b, c, 1982; Sinha, 1981; Honegger et al., 1982; Hakim Rai, 1982; Andrews-Speed and Brookfield, 1982; Baud et al., 1982). A summary description has been given in a preceding paper on palaeomagnetism of the Ladakh Intrusives (Klootwijk et al., 1979). We confine ourselves here to an overview of the geological setting of the sampled region and to selected aspects of the regional geology relevant to the present study.

The structural trends of central and eastern Ladakh follow largely the ENE–WSW trending Indus–Tsangpo suture zone. Several structural entities have been distinguished (Figs. 1–3). From north to south these are as follows:

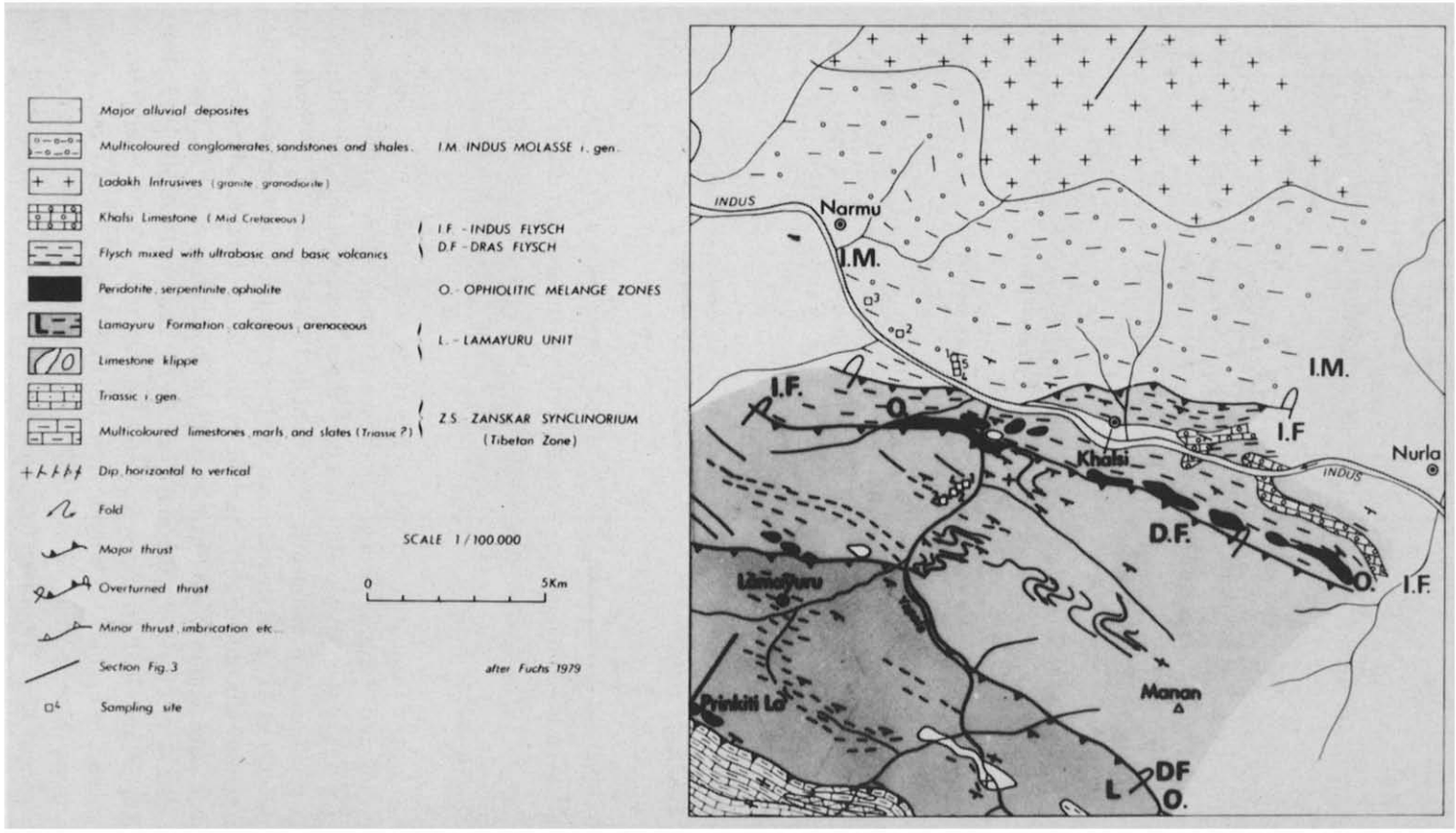


Fig. 2. Geological map of the sampled region after Fuchs (1979).

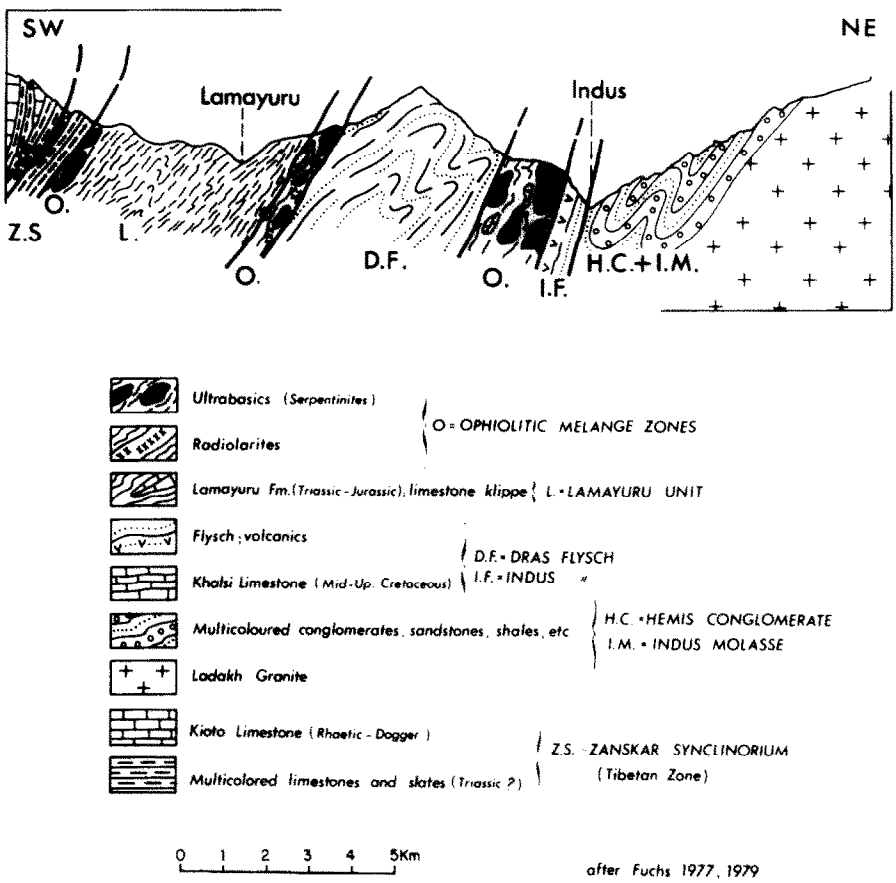


Fig. 3. Profile over the sampled region after Fuchs (1977, 1979) with slight modifications.

(1) Ladakh Intrusives

This vast belt of intrusives, situated largely north of the Indus river, ranges from olivine–norite and gabbro through diorites and tonalites to granodiorites and diorites. The age of this igneous suite has been the subject of discussion. The Ladakh Intrusives result from prolonged igneous activity (Shah et al., 1976; Frank et al., 1977; Honegger et al., 1982) which started already in pre-Late Cretaceous time, as the intrusives are unconformably overlain by the Indus Molasse, whose basal beds are dated as Late Aptian (Ravi Shankar et al., 1976; Shah et al., 1976; Fuchs, 1977, 1979; Andrieux et al., 1977; Sharma et al., 1978; Bassoullet et al., 1978, 1980a). U/Pb, Rb/Sr and Rb/Rb measurements (Honegger et al., 1982; Schärer, pers. commun., 1982) indicated ages ranging from 100 to 60 m.y. $^{40}K/^{36}Ar$ and $^{40}Ar/^{39}Ar$ determinations (Saxena and Miller, 1972; G.S.I., 1976, Le Fort, pers. commun., 1977; Sharma et al., 1978; Brookfield and Reynolds, 1981; Honegger et al., 1982)

TABLE I

Mean palaeomagnetic results Indus Molasse, Dras Flyschoids, Khalsi–Lamayura Region (34.35°N, 76.90°E), Ladakh

Formation	Direction		K	α_{95} (°)	N	Polarity N/R	Pole position (south poles)				Palaeolat.	No. (Fig. 5)
	decl. (°)	incl. (°)					lat. (°S)	long. (°E)	dp (°)	dm (°)		
(A) *Recent field components* (IM 2, DR 2)												
Indus Molasse	347.5	+51	15	4	91	N	79	157	3.5	5.5	32 N	1
Dras Flyschoids *	11	+47	–	–	–		79	17	–	–	28.5 N	2
(B) *Early Tertiary component* (IM 3, DR 3)												
Indus Molasse (IM 3)												
SITE 1	317.5	+9	7.5	16.5	13	5/8	40.5	139.5	8.5	16.5	4.5 N	3
SITE 2	306.5	+3.5	5.5	24	9	N	30.5	145.5	12	24	2 N	4
SITE 3	(347.5	+24	23	26.5	3	R)						
SITE 4	0.5	+4.5	15	10	16	N	58	76.5	5	10	2 N	5
SITE 5	350	−0.5	15.5	9	18	N	54.5	94	4.5	9	0	6
Dras Flyschoids (DR 3)												
SITE 1	357.5	+2.5	27.5	5.5	27	N	57	81.5	2.5	5.5	1.5 N	7
SITE 2	245	+0°	24.5	6.5	21	N	48.5	38	3.5	6.5	0	8
SITE 3	350	+2.5	37.5	4.5	31	N	55.5	91.5	2	4.5	1.5 N	9
SITE 4	14.5	+1.5	38	5	23	N	53.5	52	2.5	5	0.5 N	10
(C) *Possible primary component* (DR 4)												
DR 4 *	64	−32	–	–	–		10	16	–	–	17.5 S	11

* Determined from frequency distribution Fig. 4A, B.

and another Rb/Sr determination (Desio and Zanettin, 1970), in contrast, show younger ages ranging from 50 to 28 m.y.

The Ladakh Intrusives form part of an Andean-type batholith (Bassoullet et al., 1982; Honegger et al., 1982), whose nature is further illucidated by finding of inherited zircons, of an unspecified possibly Early Paleozoic age, in a leukogranite near Leh which is dated at about 63 m.y. (Schärer, pers. commun., 1982). The composition and structure of the Ladakh Intrusives is comparable with batholiths of the Gangdeze belt in southern Tibet, one of the few other geologically better explored regions of the ITS. Here, the batholiths are overlain likewise by Albo-Aptian to Lower Tertiary sediments of the Shigatze Group (Colchen, pers. commun., 1981; Kauffman, 1981), and show Rb/Sr ages ranging from 120 to 70 m.y. and from 50 to 10 m.y. (Zhang Yu-quan et al., 1981a, b; Tu Kuong-chi et al., 1981; Wang Jun-wen et al., 1981; U–Pb Dating Group, 1981; Maluski et al., 1982). The younger radiometric ages of the Ladakh Intrusives therefore may be taken as representative, either for the younger intrusive bodies, or for a resetting of isotopic systems associated with the India–Asia collision process.

(2) The Indus Molasse

The Ladakh Intrusives are overlain unconformably by the Indus Molasse, whose stratigraphy was described recently in some detail for a region in east Ladakh, east of the Zanskar River (Baud et al., 1982). In the sampled region in central Ladakh, the basal part of the Indus Molasse is made up of pebble beds, sandstone and shale, superseded by a thick Upper Cretaceous succession of multicoloured sandy shale, sandstone and conglomerate, and capped by Middle Eocene Nummulitic limestone (Ravi Shankar et al., 1976; Pal and Mathur, 1977; Bassoullet et al., 1980a). The topmost beds of what in some parts of the belt has been identified as Indus Molasse (Kargil Molasse), are even of Mio-Pliocene age (Tewari, 1964; Dixit et al., 1971; Sharma, 1979). Some authors (Bassoullet et al., 1978, 1982) consider fossiliferous limestones of Late Aptian to Albian age, outcropping along the Indus River near Khalsi, to represent lower levels of the Indus Molasse (Indus Formations in their terminology). In contrasts, Fuchs (1977, 1979) figures the tectonic contact between these limestones (Indus Flysch in his terminology) and shales of the Indus Molasse to be of prime importance as discussed in the following (3).

Sampling was restricted to the multicoloured Upper Cretaceous succession. Five sites were sampled at three localities north of the Indus river, several kilometers west of the bridge on the Khalsi–Lamayuru road (Figs. 1–3). Greenish flyschoid sandstone and red siltstone were sampled from steeply southwards dipping beds (N105°–125°E, 94°–118°S, Table I), whose top and bottom features indicate an overturned position.

(3) Dras Flyschoids and Indus Flysch

Within this flysch belt Fuchs (1977, 1979) distinguishes a zone of Indus Flysch s.s. in the north (Khalsi area and further eastwards), separated by a tectonic zone marked by ophiolitic melange from the Dras Flyschoids–Dras Volcanics zone to the south. The highly tectonized Indus Flysch s.s. consists of flyschoid shale and sandstone, volcanic material and a characteristic (Khalsi) limestone of Aptian to Cenomanian age (Gupta and Kumar, 1975; Bassoullet et al., 1978, 1982; Fuchs, 1979).

The Dras Flyschoids (Nindam Formation according to Bassoulet et al., 1978, 1980, 1982) predominate in the eastern part of the Indus basin and represent the lateral facies equivalent of the andesitic to basaltic Dras Volcanics in the west. The formation is made up of graywacke and silicified shale in a tectonized setting. Faunal content indicates a Middle to Late Cretaceous age (Ravi Shankar et al., 1976; Frank et al., 1977; Fuchs, 1977, 1979; Shah and Gergan, 1978; Bassoulet et al., 1982).

Greenish and purple coloured siliceous shale of tuffitic appearance were sampled in steeply southwestwards dipping beds (N138°–143°E, 108.5°–129°SW) at four sites along the Lamayuru–Khalsi road (Figs. 1–3). Stratigraphic facing of the sampled beds could not be established for absence of sedimentary structures, but the wider local tectonic setting (Andrieux et al., 1977; Fuchs, 1977, 1979; Bassoulet et al., 1982) indicates an overturned position (Fig. 3).

(4, 5) Lamayuru Flysch, Tethyan Himalaya

The Dras Flyschoids–Dras Volcanics belt is bounded by ophiolitic melange zones with glaucophane schists (Frank et al., 1977; Fuchs et al., 1977; 1979; Virdi et al., 1977), which separate this belt from the Ladakh Intrusives–Indus Molasse–Indus Flysch succession in the north and from the Lamayuru Flysch in the south (Figs. 2

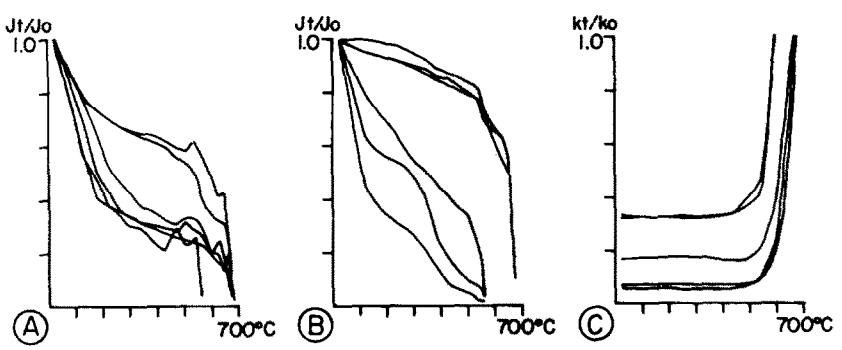
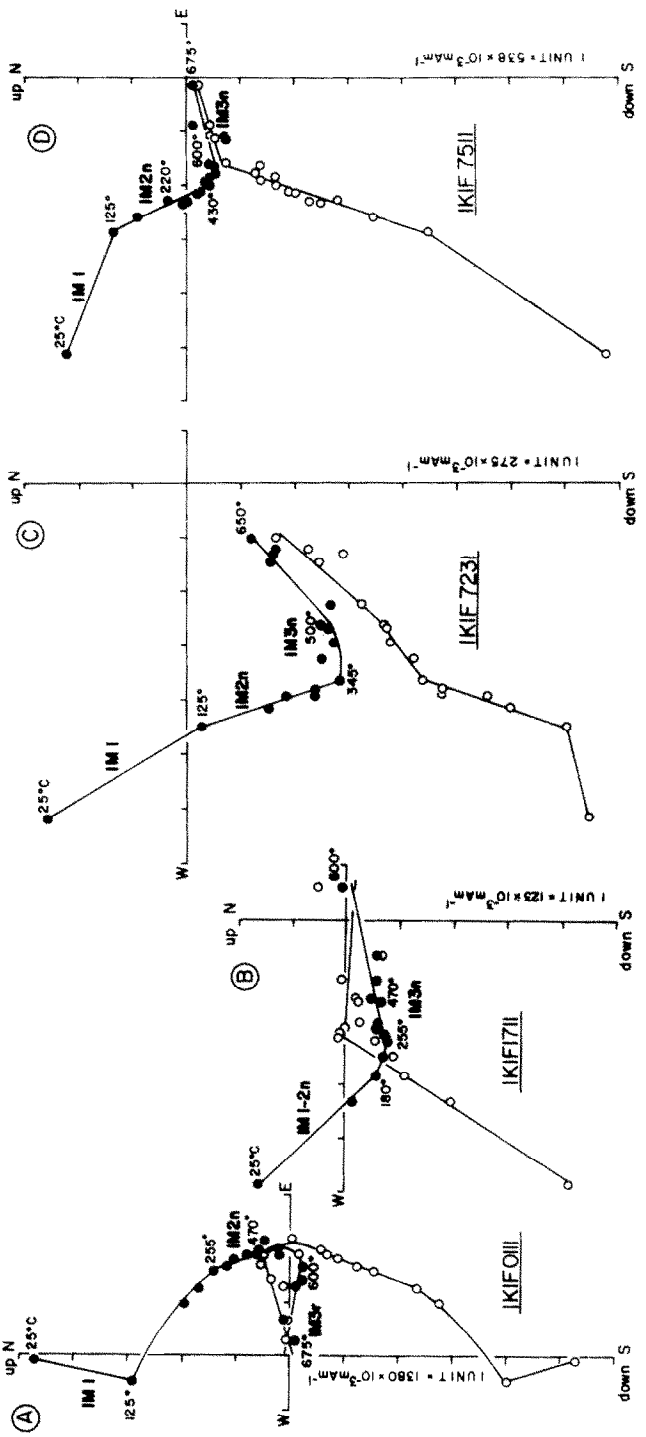


Fig. 4. Representative normalized curves showing decay of remanent magnetization (A, B) and changes in magnetic susceptibility (C) during thermal treatment.


A
B
C
D
up N
down S
W
E
25°C
IM 1
125°
IM2n
255°
470°
600°
675°
IM3n
IM 1-2n
180°
800°
220°
430°
500°
650°
345°
IKIF701
IKIF171
IKIF723
IKIF751
1 UNIT = 1380 × 10⁻⁹ mAm⁻¹
1 UNIT = 125 × 10⁻⁹ mAm⁻¹
1 UNIT = 273 × 10⁻⁹ mAm⁻¹
1 UNIT = 538 × 10⁻⁹ mAm⁻¹

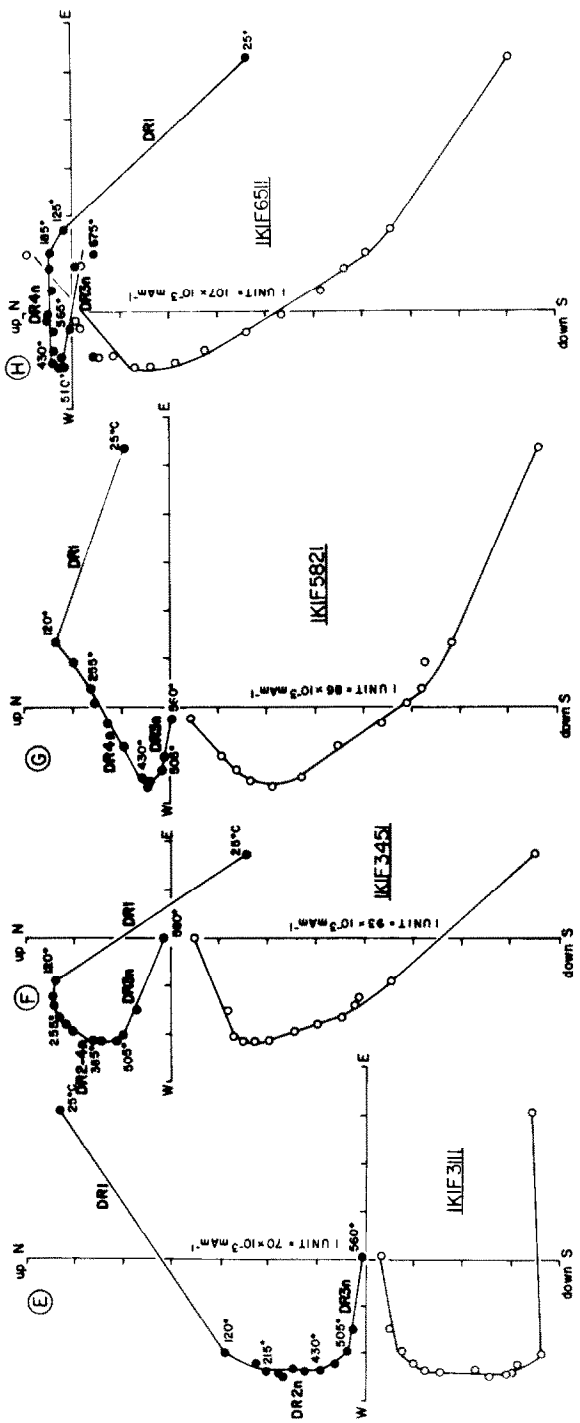
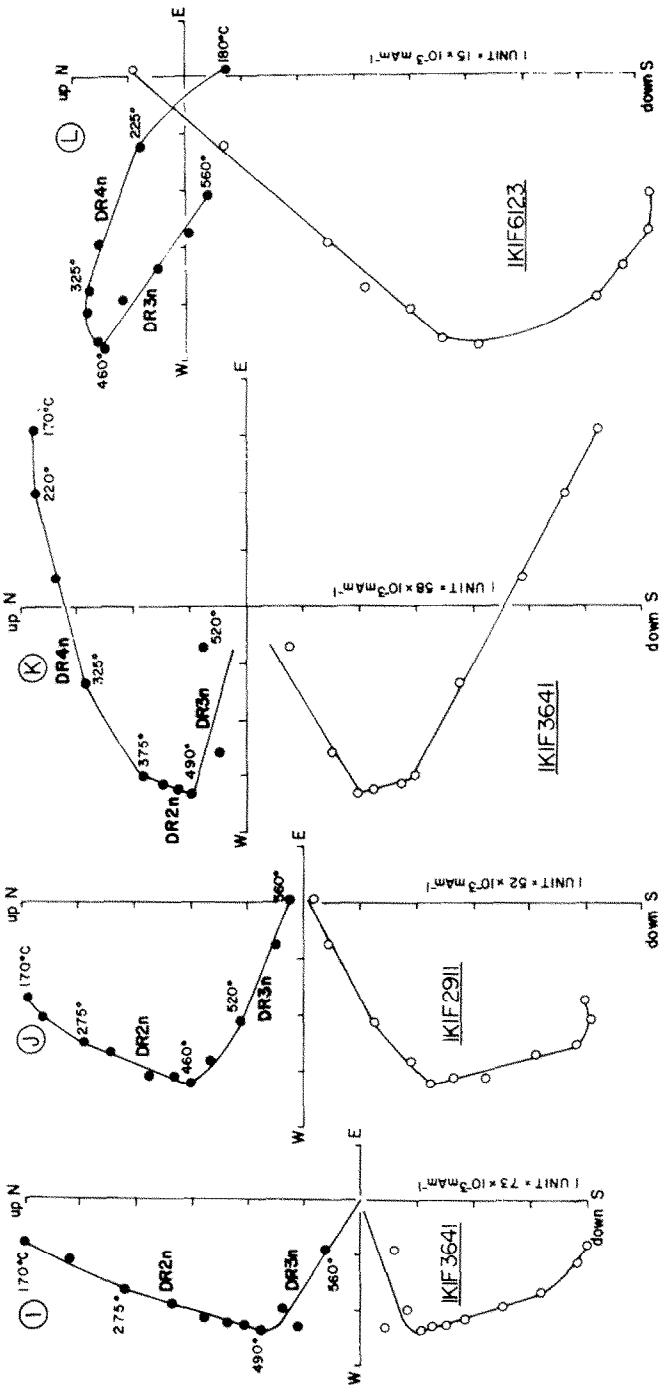


Fig. 5A–H. For legend see p. 227.


I
J
K
L
170°C
275°
DR2n
490°
DR3n
560°
IKIF3641
1 UNIT = 73 × 10⁻³ mAm⁻¹
IKIF2911
1 UNIT = 52 × 10⁻³ mAm⁻¹
220°
325°
375°
DR4n
DR2n
490°
520°
DR3n
IKIF3641
1 UNIT = 58 × 10⁻³ mAm⁻¹
225°
325°
460°
DR4n
DR3n
560°
180°
IKIF6123
1 UNIT = 15 × 10⁻³ mAm⁻¹
up N
down S
W
E

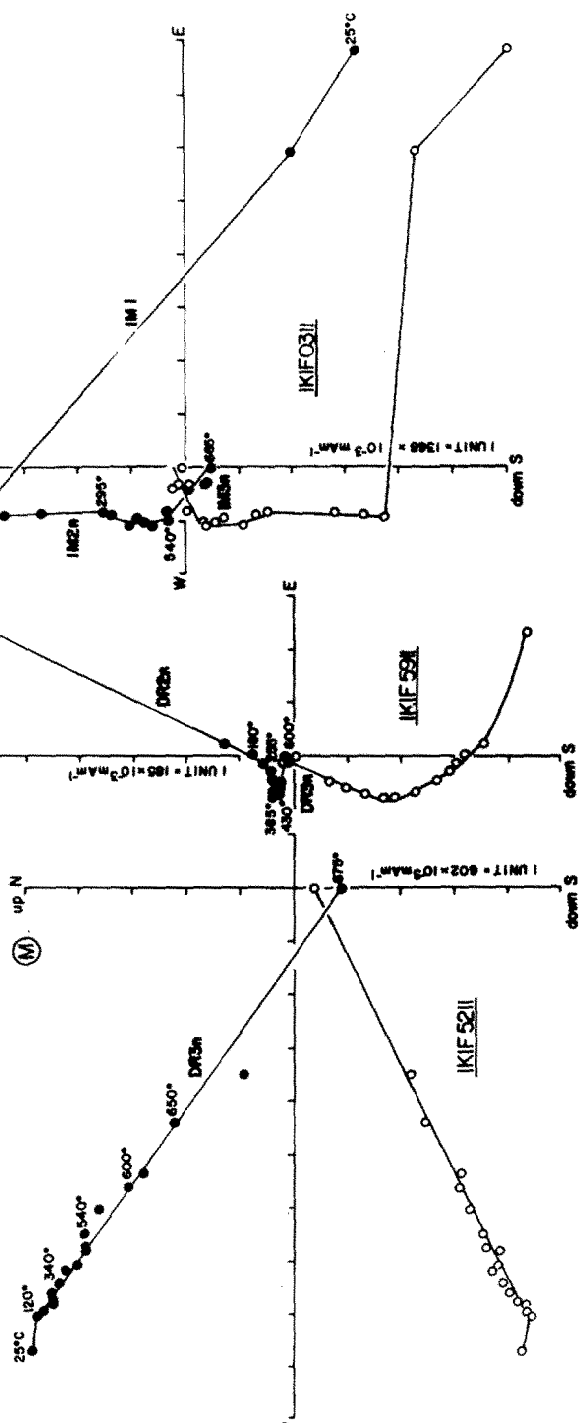


Fig. 5. Demagnetization diagrams of some representative specimens. The points denote successive positions, in orthogonal projection, of the end points of the resultant magnetization vector during progressive thermal demagnetization. Open circles denote projections on the vertical east–west plane, dots denote projections on the horizontal plane. Numbers denote successive peak temperature values. *IM1*, *DR1* = laboratory induced component; *IM2*, *DR2* = recent field component; *IM3*, *DR3* = characteristic component (secondary origin); *DR4* = component of uncertain origin, possibly primary. *n* = normal polarity, *r* = reversed polarity. Results are not corrected for bedding.

and 3). The latter flysch of Ladinian to Middle Jurassic age (Bassoullet et al., 1980a, 1981) has overthrusted Palaeozoic and Mesozoic platform carbonates of the Zanskar Range (Figs. 1–3) which in turn have overthrusted and form part of the epicontinental Tethyan-facing sedimentary cover of Greater India's northern margin.

The ITS rocks have undergone low-grade southwards increasing regional metamorphism (Fuchs, 1977; Frank et al., 1977; Gogte et al., 1981; Honegger et al., 1982), reaching prehnite–pumpellyite facies in the Dras Volcanics–Dras Flyschoids (Bassoullet et al., 1982; Honegger et al., 1982), and show superimposed schistosities (Andrieux et al., 1977; Bassoullet et al., 1980a, 1982). Bassoullet et al. (1982) distinguish four tectonic phases of alternating southern vergence and northern retrocharriage in the Lamayuru Flysch and the Zanskar Nappes. Two phases only are found in the Dras Flyschoids and the Indus Molasse. The Dras Volcanics–Dras Flyschoids have undergone a first phase of southwards directed transport which affected also the Lamayuru Flysch and the Zanskar Nappes. Its timing is disputed: post-Maastrichtian according to Fuchs (1979), earlier according to others (i.e., Bassoullet et al., 1978—pre-Cenomanian?, 1979—Late Cretaceous; Brookfield and Reynolds, 1981—prior to 83 m.y.; Frank et al., 1977—pre-Eocene). A younger pre-Middle Eocene phase of northwards directed thrusting resulted in refolding of these thrusted units and formed the first structures in the Indus Molasse, which subsequently underwent a phase of cylindrical folding. A last tectonic phase troughout the ITS zone is represented by open vertical fractures.

TREATMENT, ANALYSIS AND RESULTS

A total of 81 block samples was collected from the Indus Molasse (40) and Dras Flyschoids (41), from which 244 specimens (2.5 cm diameter, 2.2 cm length) were taken in the laboratory. All samples were oriented with a magnetic and a solar compass.

Thermal demagnetization and analysis of results were carried out according to standard procedures described in detail elsewhere (Klootwijk and Bingham, 1980; Klootwijk et al., 1981). All specimens were rinsed in diluted HCl prior to measurement. Progressive thermal demagnetization was carried out in large-size furnaces (McElhinny et al., 1971) with a feed-back controlled 10-coil Helmholtz system reducing the ambient field to less than 10 gamma over the heating-cooling cycle (about 2 hrs.). Pilot specimens, at least one per sample, were demagnetized in 13–18 steps (25°–560°C or 675°C) and the remainder of the specimens in 10–16 steps (170°–560°C or 675°C) depending on their blocking temperature range. Susceptibility increased excessively after heating above 500°–600°C (Fig. 4C). An accompanying viscous magnetization component was noted, but generally did not impede natural remanent magnetization (NRM) measurements.

Measurements were taken on a two-axis ScT cryogenic magnetometer interfaced with a HP 2100 minicomputer. Individual magnetic components were visually

identified from Zijderveld (1967, 1975) plots, and their directions were computed with an adapted version of Kirschvink's (1980) principal component analysis program.

The magnetization of the specimens was well measurable with intensities ranging between 0.2 and 20 mA m^{-1} in the Indus Molasse and between 0.4 and 9 mA m^{-1} in the Dras Flyschoids. Analysis of thermal demagnetization results indicated three magnetic components in the Indus Molasse and four magnetic components in the Dras Flyschoids.

Indus Molasse

(*IM1*) A laboratory induced component of considerable intensity (about 50% of initial NRM) which was removed below 200°C (Figs. 4A, 5A–D, O).

(*IM2*) A consistently present recent field component of normal polarity, NNE-wards and downwards pointing and representing about 25 percent of initial NRM. This component was removed between 400°C and 500°–600°C (Figs. 4A, 5A–D, O).

(*IM3*) A characteristic component of both normal and reversed polarity, SSE-

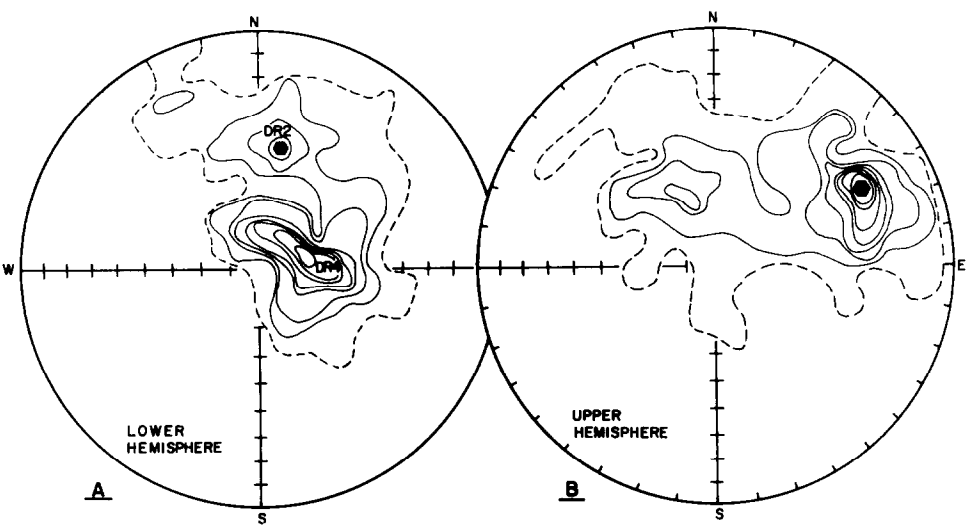


Fig. 6. A. Frequency distribution of directions for the recent field component (*DR2*, full hexagon) and the component of uncertain origin (*DR4*) in the Dras Flyschoids. Maximum grouping of *DR2* is indicated by a square. A streaking of *DR4*-component directions within a WNW–ESE oriented plane and also towards the recent field component directions is well visible. Equal area projection. Frequency interval = 2. Counting circle = 1% of hemisphere surface.

B. As above, directions corrected for bedding. Maximum grouping of DR4 component is indicated by a hexagon.

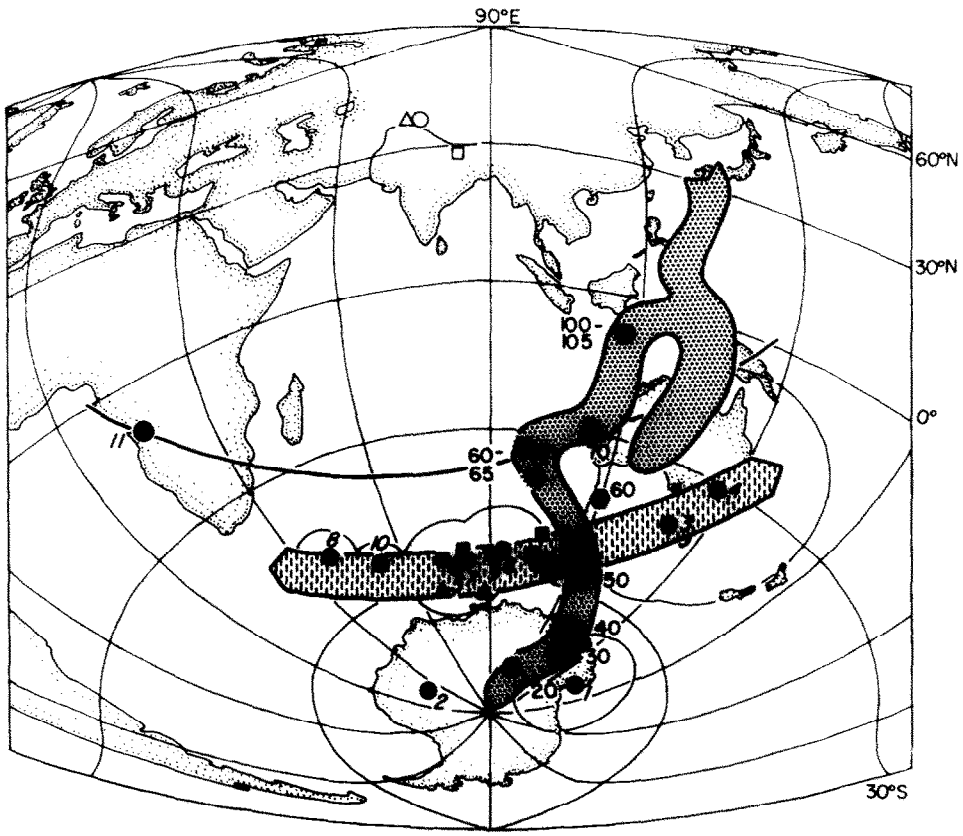


Fig. 7. Comparison of pole positions here obtained (circles: Indus Molasse (*3–6*), Dras Flyschoids (*7–10*), see Table I) with the Late Palaeozoic–Cenozoic segment of the Indo-Pakistan APWP and with pole positions for Early Tertiary secondary components observed in the Ladakh Intrusives (triangles) and in the Tibetan Sedimentary Series of central Nepal (squares). Only selected pole positions are marked on the Indo-Pakistan APWP (hexagons), i.e. for DSDP results averaged over 10 m.y. intervals (20–70 m.y.), for the Early Tertiary Deccan Traps (> 60–65 m.y.) and for the Early Cretaceous Rajmahal Traps (> 100–105 m.y.). The swath formed by the loci of pole positions for the secondary components (*3–10*) intersects the Indo-Pakistan APWP in the 50–60 m.y. interval, without account being made for post-collisional crustal shortening in Greater India's northern margin. The pole position and locus for the DR4-component (circle *11*), assuming a primary origin and not accounting for crustal shortening in Greater India's northern margin, and for the recent field components (circle *1*, *2*, Table I) are indicated also. The inverted triangle on the Indo-Pakistan APWP indicates the pole position for the Panjal Traps (McElhinny et al., 1978), interpreted as an Early Tertiary overprint by the authors.

wards and NNW-wards directed with shallow inclination, and representing about 25% of initial NRM. This component was removed either between 550°C to 600°C (Figs. 4A, 5B) or between 560°C to 675°C (Figs. 4A, 5A, C, D).

Dras Flyschoids

(*DR1*) A magnetically soft component of considerable intensity (about 40% of initial NRM), generally E to SE-wards pointing and of shallow inclination, removed below 150°C (Figs. 4B, 5E–H).

(*DR2*) A recent field component of normal polarity, NNE-wards and downwards directed and representing about 30% of initial NRM, removed between 200°C and 500°C (Fig. 4B, 5E, F, I–K, N).

(*DR3*) A characteristic component of normal polarity, northwards pointing and of shallow inclination. This component, representing less than 25% of initial NRM, was removed between 500°C to 565°C (Fig. 4B, 5E–L, N) and up to 675°C in the purple coloured silicified shale (Fig. 4B, 5M).

(*DR4*) A steeply downwards directed component, streaking in a WNW–ESE directed plane and towards the mean direction of the recent field component (Figs. 4B, 5F–H, K, L, 6). This component was broken down between 200°C and 375°–450°C, i.e. in the same temperature interval as component DR2. Curved segments, visible on several of the Zijderveld plots (Fig. 5F, K), indicate simultaneous breakdown of both components. This may account for a smeared distribution between the mean groupings of both component directions. Because of this streaking, determination of mean directions for components DR2 and DR4 from mean groupings of density distributions (Fig. 6B) seems preferable to Fischerian statistics (Table I), although confidence limits cannot be obtained this way.

INTERPRETATION

The site-mean directions for the characteristic components of Indus Molasse and Dras Flyschoids are all of very low inclination (Table I). A dispersion is notable though in the declinations, particularly for the Indus Molasse. This dispersion reflects the complex local tectonic setting. Tectonic activity continued apparently till recent time as indicated by the similarity in patterns of eastwards and westwards offsets with respect to the Indo-Pakistan APWP for characteristic (IM3, DR3) and for recent field (IM2, DR2) pole positions of the Indus Molasse and Dras Flyschoids respectively (Fig. 7). The dispersed declination pattern of the characteristic components has little interpretative value, other than exemplifying local tectonic complications. The coherent low inclinations, however, indicate a very low northern (0°–5°N, Table I) palaeolatitude for the region at the time of their acquisition.

South-pole positions for the characteristic components fall in a swath which intersects the Indo-Pakistan apparent polar wander path (APWP) in the 50–60 m.y. interval (Fig. 7). This swath contains also pole positions for secondary components observed in the Tibetan Sedimentary Series of the Thakkhola region of central Nepal which were correlated with a K–Ar overprint at 50–60 m.y. ago (Klootwijk and Bingham, 1980), and pole positions from the Ladakh Intrusives which were interpre-

ted as secondary components associated with closure of the isotopic systems at about 50 m.y. ago (Klootwijk et al., 1979). In view of the regional subgreenschist metamorphism of the sampled region and the close similarity of the here obtained pole positions with the loci of the above mentioned pole positions, a secondary origin is envisaged likewise for the characteristic components observed in the present study.

The data from the Ladakh Intrusives and the Thakkhola region have been interpreted before (Klootwijk et al., 1979; Klootwijk, 1981; Klootwijk and Radhakrishnamurty, 1981) in terms of collision of Greater India and an island arc positioned off Asia's southern margin at low northern palaeolatitudes (0°–10°N), for which an age was concluded between 50 to 60 m.y. ago. This interpretation of palaeomagnetic data was based on seafloor spreading data and the geological record of the ITS, indicating collision at about 55 m.y. ago (Powell and Conaghan, 1975; Powell, 1979). Recent indications for earlier tectonic phases in the Himalayan region and in southern Tibet and for an Andean-type origin of the Ladakh Intrusives, however, warrant a more elaborate interpretation of the palaeomagnetic data.

We have to clarify that the intersection of the swath formed by the secondary pole positions from the Ladakh Intrusives, the Indus Molasse, the Dras Flyschoids and the Tibetan Sedimentary Series of the Thakkhola region with the Indo-Pakistan APWP in the 50–60 m.y. interval, does not necessarily reflect the actual timing of collision as suggested before (Klootwijk et al., 1979; Klootwijk, 1981; Klootwijk and Radhakrishnamurty, 1981). Alternatively, this intersection may date either a phase of enhanced thermo-chemical activity associated with the collision process or a late phase of relaxation in the thermal and/or stress regime, postdating actual collision. The palaeomagnetic data effectively place only a Late Palaeocene–Early Eocene (50–60 m.y.) upper time limit on collision, which accords with finding of a strong Asiatic affinity of Indian mammals since Late Palaeocene time (Sahni et al., 1981). These data have no direct bearing on the timing of the suggested Late Cretaceous obduction of an island arc (Kohistan Complex/Dras Volcanics–Dras Flyschoids) onto Greater India's northern margin (Tapponnier et al., 1981a, b; Brookfield and Reynolds, 1981; Mercier and Tapponnier, 1981), when Greater India was still at moderately southern palaeolatitudes.

An alternative interpretation for the DR4-component in the Dras Flyschoids (Table I) is interesting in this respect. The magnetically rather soft nature of part of the observed components and the streaking of directions both in a WNW–ESE directed plane and towards the recent field direction (DR2) suggest that this component may represent a low-temperature oxidation phase of recent origin, with a direction aligned preferably into and towards the predominant foliation plane. Such an interpretation follows similar observations in the Ladakh Intrusives (Klootwijk et al., 1979), the Thakkhola region (Klootwijk and Bingham, 1980), Kashmir (Klootwijk et al., 1983) and the Krol Belt (Klootwijk et al., 1982a), but is not entirely satisfactory in the present case. The predominant foliation plane dips steeply southwards (Andrieux et al., 1977; Bassoullet et al., 1978) and not northwards as

would be expected according to the DR-4 component (Fig. 6A).

Alternatively, a primary origin may be advanced for this component, in which case the intersection of the locus of its pole position with the Late Cretaceous segment of the Indo-Pakistan APWP (Fig. 7) supports the hypothesis of Late Cretaceous collision and obduction at moderately southern palaeolatitudes. If this interpretation is to be accepted, a viable explanation will be needed for preservation of a magnetically softer remanence (DR4) under conditions (thermo-chemical activity?) leading to subsequent acquisition of a harder component (DR3).

CONCLUSIONS

Characteristic magnetization components observed in Upper Cretaceous Indus Molasse and in Middle to Upper Cretaceous Dras Flyschoids from the Indus–Tsangpo suture zone in Ladakh (northwestern Himalaya) indicate an equatorial to low northern palaeolatitude of acquisition. This is similar to palaeolatitudes indicated by secondary magnetization components observed in the Ladakh Intrusives (Klootwijk et al., 1979) and in the Tibetan Sedimentary Series from central Nepal (Klootwijk and Bingham, 1980). The former characteristic components are interpreted likewise as a secondary magnetization acquired between 50 and 60 m.y. ago during collision of Greater India with Asia's southern margin. The latter secondary magnetizations have been interpreted before in terms of constraints on palaeolatitude and time of collision of Greater India with an island arc off southern Asia. Recent geological indications for a Late Cretaceous tectonic phase in southern Tibet and the Himalayan region and an Andean-type origin for the Ladakh Intrusives, urge a reconsideration of this interpretation. In fact, the secondary magnetization components may represent only an Early Tertiary phase of relaxation in temperature and stress conditions. If so, the suggested Late Cretaceous obduction of an ophiolite-island arc complex (Kohistan Complex/Dras Volcanics–Dras Flyschoids) onto Greater India's northern margin cannot be detailed in terms of time and palaeolatitudinal position from these secondary components. A persistently present but rather soft magnetic component (DR4) in the Dras Flyschoids, however, may be interpreted in support of this hypothesis. When taken as a primary magnetization, this component indicates an about 20° southern palaeolatitude during the Middle to Late Cretaceous for the Dras Flyschoids, which would support the late Cretaceous obduction model.

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